

LabLens — Feature List

Columbia × NYU Claude Builder Hackathon

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1 Product Overview

LabLens is an AI-powered research alignment engine that matches university students to professors whose active, funded research aligns with their intellectual interests. It goes beyond search or directory lookup — it is an intelligence layer that produces actionable research guidance, personalized outreach, and privacy-preserving matching.

2 Core ML Pipeline

1. Intent Extraction (Claude)

- Parses a student’s free-text research interest into structured fields: primary domain, specific topics, methodological interests, career direction, search keywords, and an embedding-optimized description.
- Adapts extraction based on student level (undergrad, master’s, PhD applicant).

2. Live Faculty Discovery (Linkup API)

- Multi-query agentic search to find professors at a given university in the student’s research domain using live web data.
- Deep profile retrieval per professor: lab website, 5 most recent papers (2023–2025) with abstracts, active NSF/NIH grants, current PhD students and postdocs, recent graduates and their next positions, and open recruiting announcements.
- Lab alumni discovery via LinkedIn intelligence (PhD graduates, current positions, research topics).

3. Semantic Embedding and Cosine Similarity Scoring

- Uses OpenAI `text-embedding-3-small` (1536-dimensional vectors) to embed both the student’s interest and each professor’s paper abstracts and research statement.
- Computes cosine similarity between the student vector and each paper vector.
- Weighted scoring: 70% from paper similarity (blend of max and average across up to 5 recent papers), 30% from research statement similarity.
- Normalizes raw cosine (typically 0.1–0.9) to a 0–100 alignment score.
- Reports per-paper scores, best-matching paper index, and confidence level (high / medium / low based on number of papers embedded).
- Ranks all candidate professors by alignment score descending.

4. Claude AI Synthesis Layer

- Generates all qualitative intelligence via Anthropic Claude Sonnet, including professor cards, seed ideas, emails, skills gap analysis, lab culture reports, and timing assessments.

3 Feature Modules

3.1 Professor Intelligence Card

- Plain-language summary of the professor’s *current* research focus (not career history).
- Three specific open research questions the professor is actively investigating.
- One-sentence explanation of why the student’s interest aligns.
- Best paper to read first, with a reason why.

3.2 Semantic Alignment Score and ML Explainability Panel

- Animated alignment score badge per professor (Strong Match ≥ 85 , Good Match 70–84, Partial Match < 70).
- Animated alignment bar on each professor card.
- Expandable “How we scored this” panel showing: model used, method (cosine similarity over research embeddings), raw cosine value, normalized score, and per-paper score breakdown with visual bars.
- Explicitly communicates that the score is a real ML computation, not an LLM opinion.

3.3 Personalized Seed Research Ideas

- Three original research seed ideas per professor, each sitting at the intersection of the student’s interest and the professor’s open questions.
- Each idea includes: a title, a specific research question, connection to the professor’s work, connection to the student’s interest, and a difficulty rating (undergrad / master’s / PhD level).
- Ideas are grounded in the professor’s actual papers — never generic.

3.4 Lab Culture Health Report

- Culture score (0–100) derived from publicly available signals.
- Evidence-based strengths and neutral-language watch-fors.
- Typical graduation timeline based on alumni data.
- Mentorship style signals inferred from public data.
- “Best fit for” student profile description.
- Three suggested questions to ask in an interview.
- Data sources: alumni time-to-graduation, student first-authorship rate, alumni destination diversity, lab size, public engagement, and public mentions.

3.5 Lab Alumni Trail

- Tracks where past PhD graduates and postdocs went after leaving the lab.
- Shows each alumnus’s name, graduation year, current role, employer, and research topics.
- Summary statistics: industry vs. academia placement count, average time in lab.

3.6 Timing Intelligence Engine

- Determines the optimal timing for a student to contact a professor.
- Inputs: current date, semester and week, proximity to grant deadlines, grant deadline type, days since last arXiv submission, upcoming conference deadlines, and whether it is summer.
- Outputs: timing score (0–100), verdict (excellent / good / caution / wait) with color coding, primary reason, detailed explanation, specific optimal send time (e.g., “Tuesday April 14 at 7:35 AM”), next contact window if verdict is “wait”, and timing red flags.
- Calendar visualization showing semester position, grant deadlines, conference deadlines, and highlighted optimal contact windows.

3.7 Skills Gap Analysis

- Compares the student’s background against the professor’s lab requirements (methods, programming tools, mathematical frameworks from papers).
- Lists required skills with importance level (required / helpful / bonus) and whether the student already has each.
- Identifies critical gaps and existing strengths.
- Provides a two-week prep plan with specific daily learning tasks and free resource links.
- Suggests how to frame the student’s background in the email to minimize gap concerns.

3.8 Grant Opportunity Matching

- Matches relevant fellowships (NSF GRFP, Hertz, Ford, etc.) to the student.
- Each match includes: name, amount, deadline, fit score, and “Why this fits you” text.
- Shows the professor’s active grants that fund RAs (with amounts from NSF Awards API).
- Highlights when applying to a matched grant would strengthen an RA candidacy.

3.9 Cold Email Generator

- Generates a personalized cold email draft per professor.
- Subject line references a specific paper title — never generic.
- Opening shows the student read the paper (references a specific finding or method).
- Body connects the student’s background to a seed idea as an intellectual hook.
- Ask offers both a 15-minute call and email reply to reduce friction.
- Hard-capped at 180 words, enforced programmatically.
- Avoids cliché phrases (“I am reaching out”, “Hope this finds you well”).
- Never starts with “I” — starts with something about the professor’s work.
- Tone: peer-to-peer curiosity, not job application.
- Copy-to-clipboard button and “Edit & Personalize” toggle.

3.10 Follow-Up Tracker

- State machine: SENT $\rightarrow$ WAITING $\rightarrow$ FOLLOW_UP_READY $\rightarrow$ FOLLOWED_UP $\rightarrow$ REPLIED / ARCHIVED.
- Tracks email sent date, shows countdown to follow-up window (10 days).
- Auto-generates a follow-up email draft when the window opens.

3.11 PhD Program Matching

- Separate flow triggered by “I’m applying to PhD programs” toggle.
- Ranks top 8 PhD programs using the same embedding pipeline.
- Per program: research fit score, number of active professors in the student’s area, placement rate, average time-to-graduation, top 2 professors to contact pre-application, and an application strategy note.
- Uses CSRankings data (via Linkup), NSF placement data, and semantic embeddings.

4 Novel ML Contribution: Three-Tier Privacy-Preserving Matching

4.1 Problem Statement

Professors often have confidential research projects (industry NDA, pre-publication, grant-pending, politically sensitive) that they cannot publicly describe but for which they need specific types of collaborators. No existing system enables matching across this confidentiality boundary.

4.2 Three-Tier Data Architecture

- Tier 1: Public (weight 50%)** — Published papers, abstracts, lab website, NSF/NIH grants, arXiv submissions, public engagement. Full Linkup search and full embedding.
- Tier 2: Semi-Public (weight 30%)** — Workshop papers, grant abstracts, course syllabi, GitHub repos, collaborator acknowledgments. Requires Linkup deep search to surface.
- Tier 3: Private / Confidential (weight 20%)** — Industry-funded projects, pre-submission work, unawarded grant proposals, sensitive topics. Professor inputs only “what type of mind I need” (skills and intellectual approach), *not* what the project is. A differentially private embedding is generated and the raw text is immediately and irreversibly discarded.

4.3 Differential Privacy Mechanism

- Gaussian noise injection calibrated to a privacy budget ϵ (default $\epsilon = 1.0$; options: 0.5 high privacy, 1.0 balanced, 2.0 lower privacy / higher accuracy).
- L2 sensitivity of unit-normalized embeddings bounded by 2.
- Noise scale = $2/\epsilon$; noised embedding re-normalized to the unit sphere to preserve angular relationships for cosine similarity.
- Raw text is never stored, never logged, never returned — only the noised vector persists.
- Privacy middleware blocks request body logging on the private signal endpoint.

4.4 Three-Tier Composite Scoring

- When a private signal exists: $\text{composite} = 0.50 \times T1 + 0.30 \times T2 + 0.20 \times T3$.
- When no private signal: weights renormalize to $0.625 \times T1 + 0.375 \times T2$.
- Agreement bonus: +3 points if Tier 3 and Tier 1 scores agree within 15 points; -2 penalty otherwise (reflects noise uncertainty).
- Students see a higher alignment score with a lock badge and the message “This professor has indicated additional research interest beyond their public profile that aligns with you” — without ever seeing what was written.

4.5 Professor Portal

- Dedicated `/professor` route for professors to manage their profile.
- Three-tier profile visualizer showing data coverage across tiers.
- Private signal input form with privacy level selector and explicit “text discarded” confirmation flow.
- Professors can delete their private signal at any time.

5 User Interface and Pages

5.1 Page 1: Landing / Search

- Full-screen dark hero with animated dot grid background.
- Large textarea for natural-language interest input with rotating placeholder examples.
- University selector dropdown (Columbia, NYU, MIT, Stanford, CMU, UC Berkeley, Harvard — demonstrating generalizability).
- Student level selector (Undergrad / Master’s / PhD Applicant).
- Feature badges: Live research data, ML alignment scoring, Linkup Search, Claude AI.

5.2 Page 2: Results

- Left sidebar for filters and search refinement.
- Professor cards in a 2-column grid.
- Sort options: Best Match / Most Approachable / Newest Hire.
- Filter pills by department.
- Each card shows: alignment score badge, timing badge, current focus summary, animated alignment bar, seed idea count, alumni count, timing verdict, and grant count.

5.3 Page 3: Professor Deep Dive

- Six tabbed sections, each lazy-loaded: (1) Research Intelligence — full card, seed ideas, best paper, ML score breakdown; (2) Lab Intelligence — culture report and alumni trail side by side; (3) Timing & Approach — timing score, calendar visualization, optimal send time; (4) Skills Gap — lab requirements vs. student skills, two-week prep plan; (5) Grant Opportunities — matched fellowships and professor’s active grants; (6) Your Email — generated email, copy button, edit mode, follow-up tracker.

5.4 Page 4: PhD Program Matching

- Top 8 programs ranked by embedding-based research fit score.
- Per-program details and professor contact strategy links.

5.5 Page 5: Professor Portal

- Three-tier data visualizer and private signal management interface.

6 Loading, Error Handling, and UX Polish

- Five-step loading sequence with progress descriptions (searching faculty data, reading publications, computing alignment, analyzing culture, generating roadmap).
- Skeleton loaders with shimmer animation (no spinners).
- Graceful degradation: “Limited public data” badge when Linkup returns nothing; keyword-match fallback when embedding fails; refresh button when NSF API times out.
- Score numbers animate counting up on load; professor cards stagger in with 150ms delay; alignment bars animate width on mount; cards elevate with subtle glow on hover.
- Pre-loaded demo scenarios for reliable live demonstrations.
- Mobile responsive design.

7 Tech Stack Summary

- **Frontend:** Next.js 14+ with TypeScript, Tailwind CSS, Framer Motion
- **Backend:** Python FastAPI
- **ML Core:** OpenAI text-embedding-3-small for semantic embeddings; NumPy for cosine similarity and Gaussian differential privacy
- **Search:** Linkup API (deep agentic search for live professor data)
- **AI Synthesis:** Anthropic Claude Sonnet for reasoning, summarization, seed idea generation, and email drafting
- **Public Data APIs:** NSF Awards API, arXiv API, academic calendar logic
- **Database:** In-memory for hackathon (no auth)